

Bluestock Mutual Fund Analytics Capstone Project

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Organization: Bluestock Fintech

Mutual Fund Analytics Platform using Python, SQLite, Power BI and Advanced Financial Analytics:

➤ **Executive Summary :**

The Bluestock Mutual Fund Analytics Capstone Project was designed to build an end-to-end analytics platform for the Indian mutual fund industry. The project involved collecting, cleaning, storing, analyzing, and visualizing mutual fund data using Python, SQLite, SQL, Jupyter Notebook, and Power BI.

The project covers the complete analytics lifecycle, beginning with data ingestion and cleaning, followed by database creation, exploratory data analysis, performance evaluation, advanced risk analytics, and dashboard development. Key financial metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Value at Risk (VaR), and Conditional Value at Risk (CVaR) were calculated to evaluate fund performance and risk.

Interactive dashboards were developed in Power BI to visualize industry trends, fund performance, investor behavior, SIP growth, and market movements. Advanced analytics including investor cohort analysis, SIP continuity analysis, sector concentration measurement using HHI, and a rule-based fund recommender system were implemented. The final outcome is a comprehensive mutual fund analytics solution that helps investors, analysts, and fund managers make data-driven investment decisions.

➤ **Project Objectives :**

1. Build a complete ETL pipeline for mutual fund datasets.
2. Store cleaned datasets in a structured SQLite database.
3. Perform exploratory data analysis to understand market and investor trends.
4. Compute mutual fund performance metrics and benchmark comparisons.
5. Develop an interactive Power BI dashboard for business users.
6. Perform advanced risk and portfolio analytics.
7. Build a simple recommendation engine for investors.

8. Deliver professional documentation, dashboards, and reports.

➤ **Data Sources :**

The project uses multiple mutual fund datasets collected from publicly available financial sources and processed for analysis. The datasets include mutual fund master data, historical NAV records, AUM statistics, SIP inflows, category-wise inflows, investor transaction data, benchmark indices, and portfolio holdings.

The final cleaned datasets used in the project are:

1. Fund Master Data
2. NAV History Data
3. AUM by Fund House
4. Monthly SIP Inflows
5. Category Inflows
6. Industry Folio Count
7. Scheme Performance Data
8. Investor Transactions
9. Portfolio Holdings
10. Benchmark Indices

All datasets were cleaned, standardized, validated, and stored in structured formats before analysis.

ETL Design :

The ETL (Extract, Transform, Load) pipeline was developed using Python.

○ **Extract;**

Raw datasets were collected from CSV files and mutual fund data sources.

○ **Transform;**

Data cleaning operations included:

- Handling missing values
- Removing duplicate records
- Standardizing date formats

- Correcting data types
- Validating numerical fields
- Creating derived metrics

- **Load**

The cleaned datasets were stored in SQLite and exported into processed CSV files for analytics and dashboard development.

The ETL pipeline ensures consistent, accurate, and analysis-ready data across the project.

➤ **Database Design :**

SQLite was used as the primary database engine for this project.

The database stores all cleaned mutual fund datasets in structured tables. Relationships were maintained using common identifiers such as AMFI Code and Date fields.

Major tables include:

- Fund Master
- NAV History
- AUM Data
- SIP Inflows
- Category Inflows
- Investor Transactions
- Portfolio Holdings
- Benchmark Indices

SQL queries were used to retrieve, aggregate, and analyze mutual fund information efficiently.

The database design supports scalable analytics and dashboard reporting.

➤ **Exploratory Data Analysis (EDA) :**

Exploratory Data Analysis was performed to understand mutual fund industry trends, investor behaviour, fund characteristics, and transaction patterns. Multiple visualizations were generated using Python libraries including Pandas, Matplotlib, and Seaborn.

AUM Analysis;

The Assets Under Management (AUM) trend was analyzed to understand the growth of the mutual fund industry and identify leading fund houses based on assets managed.

NAV Trend Analysis;

Historical NAV movements of mutual funds were analyzed to study long-term growth patterns and market behaviour.

SIP Inflow Analysis;

Monthly SIP inflow trends were examined to understand retail investor participation and investment discipline.

Investor Demographics;

Investor behaviour was analyzed using demographic variables such as age group, gender, state, and city tier.

Category Analysis;

Fund categories were compared to identify segments attracting the highest investor interest and inflows.

Correlation Analysis;

A correlation matrix was developed to identify relationships among important financial and investment variables.

Key EDA Findings;

- Mutual fund AUM has shown a steady upward trend.
- SIP inflows increased consistently, indicating growing investor participation.
- Equity-oriented schemes attracted the majority of investments.
- Tier-1 cities contributed the highest transaction volume.
- Younger investors showed higher participation in SIP investments.

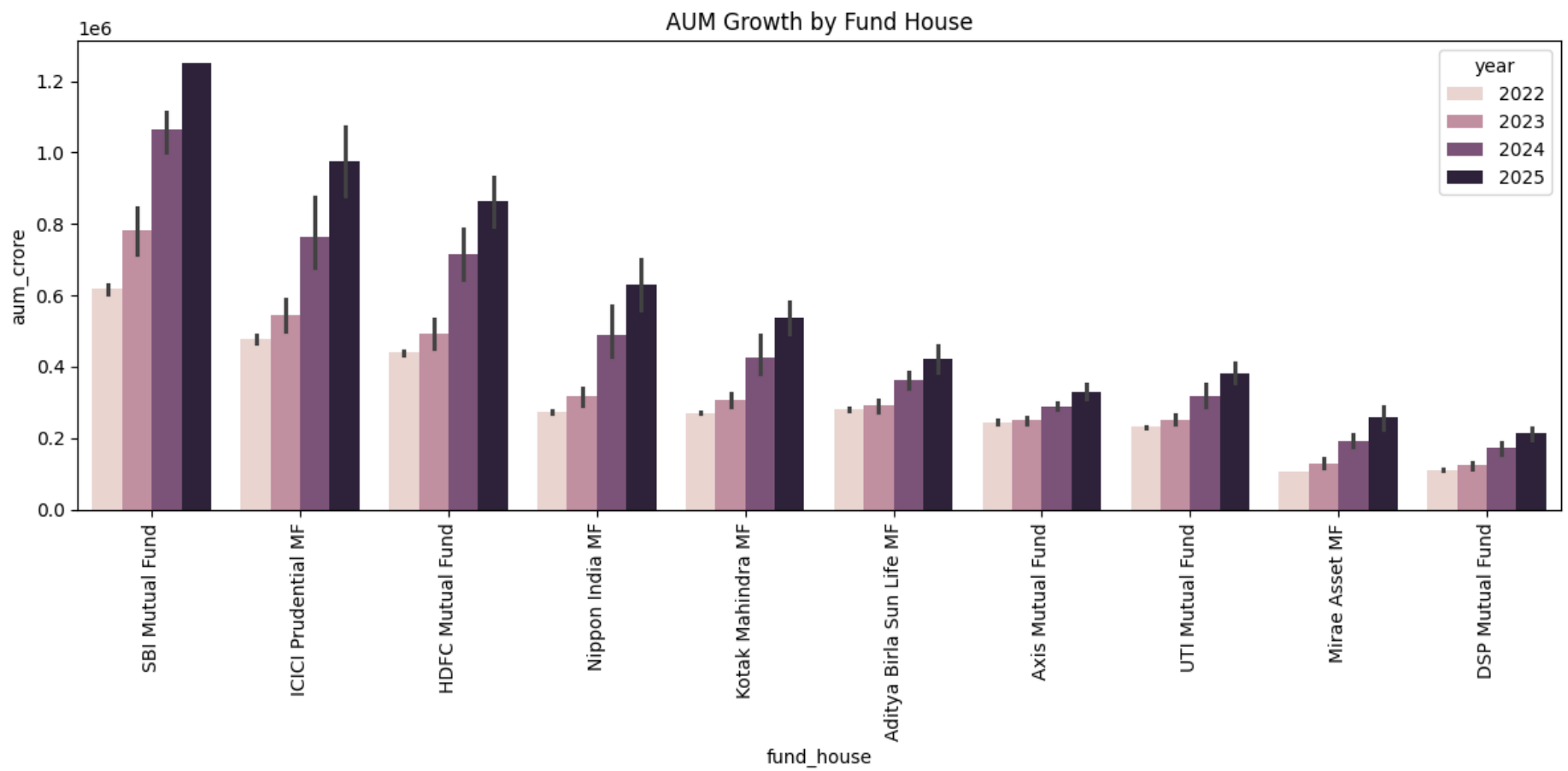


Figure 1: AUM Growth Analysis

Daily NAV Trend of All 40 Mutual Fund Schemes (2022-2025)

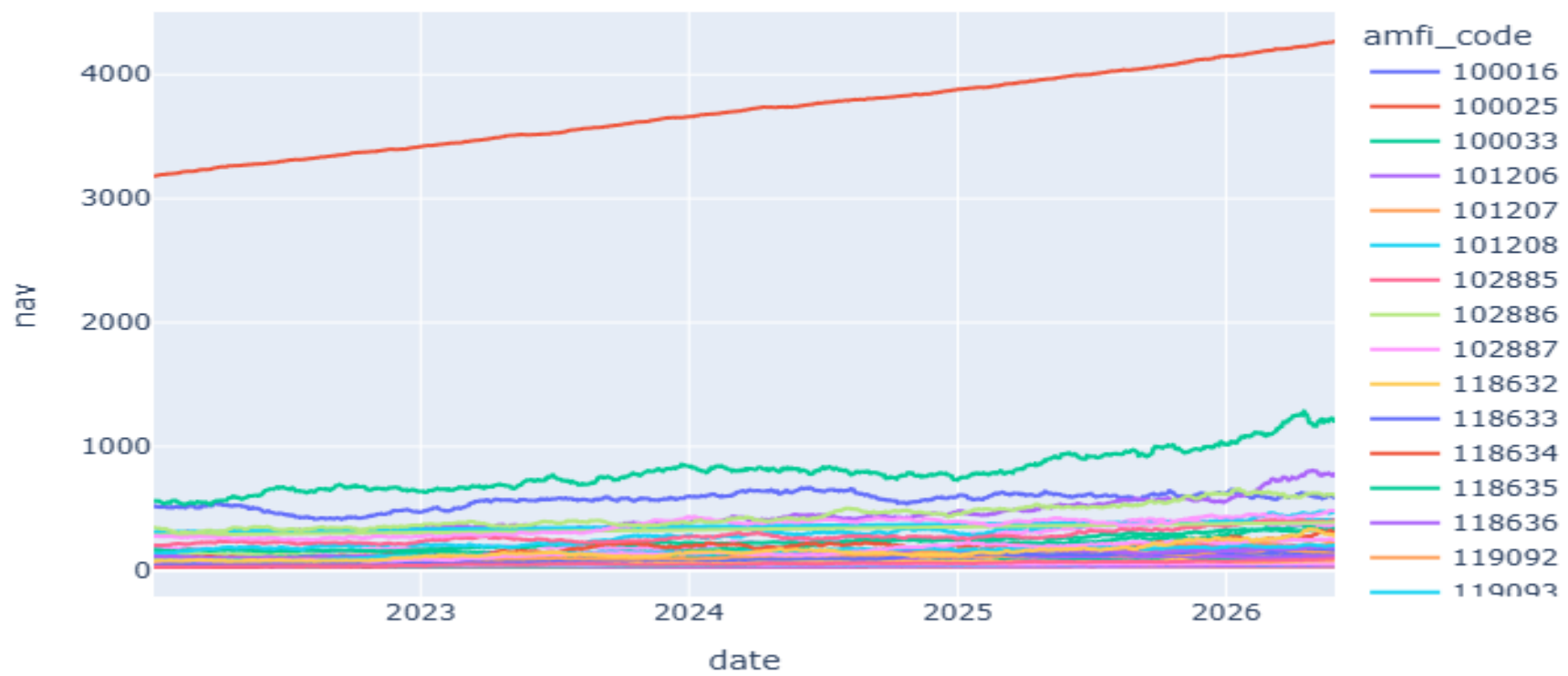


Figure 2: NAV Trend Analysis

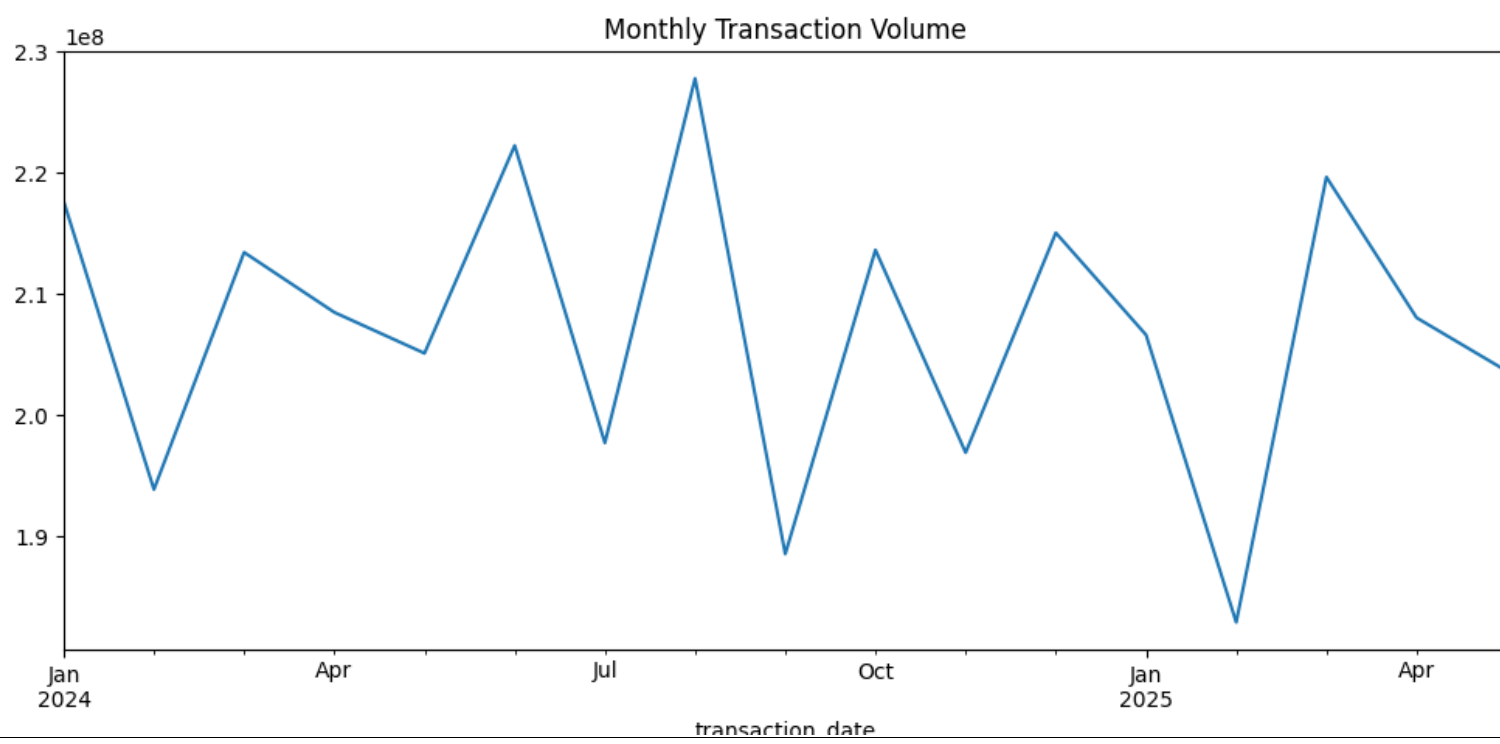


Figure 3: Monthly Transaction Volume

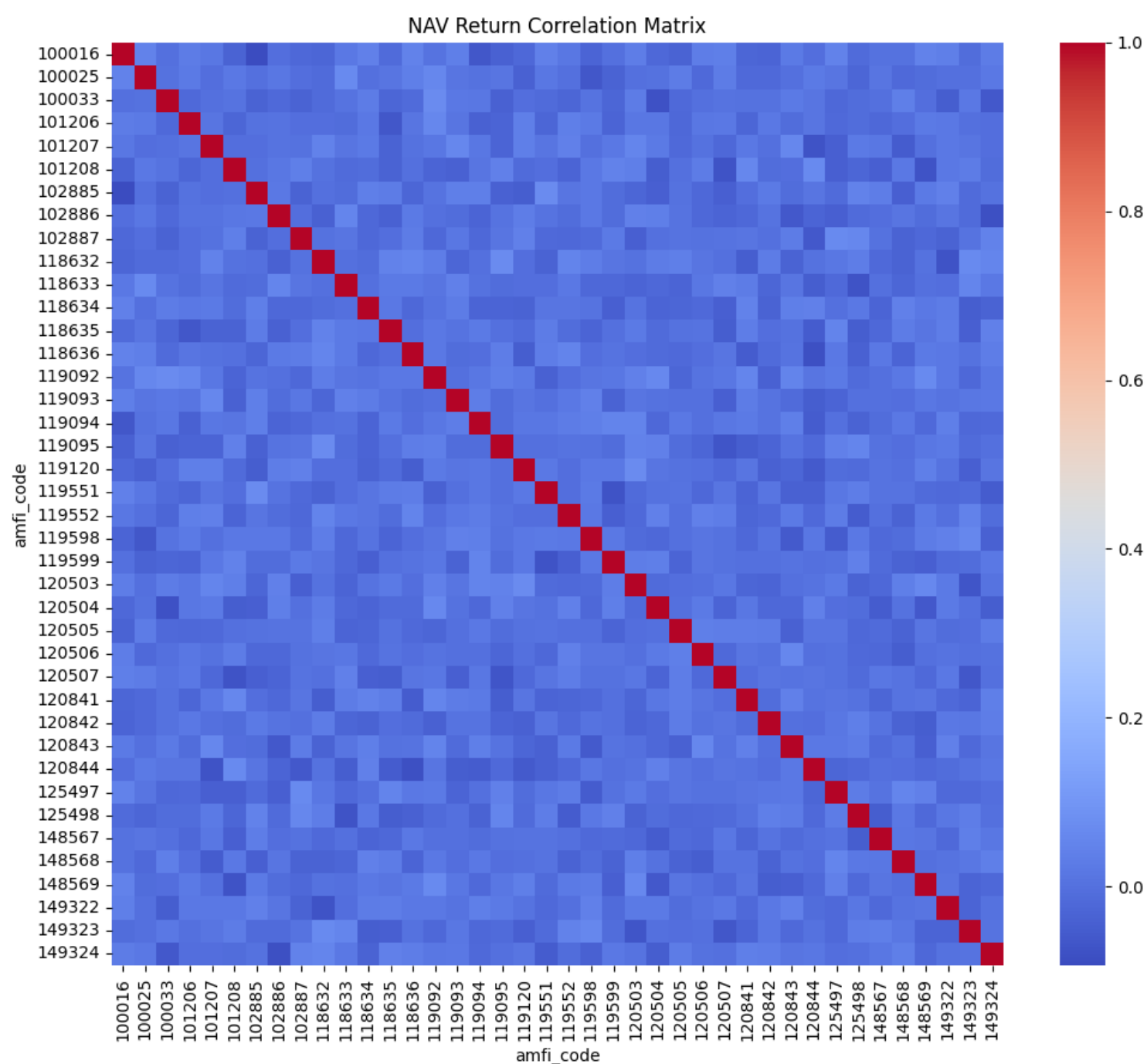


Figure 4: Correlation Matrix

➤ Performance Analytics :

Performance analytics were conducted to evaluate returns, risk, and benchmark performance of mutual funds.

CAGR Analysis

Compound Annual Growth Rate (CAGR) was calculated for 1-year, 3-year, and 5-year periods. CAGR helps measure the annualized growth rate of a fund over a specific investment horizon.

Sharpe Ratio

Sharpe Ratio measures risk-adjusted returns by comparing excess returns over the risk-free rate with portfolio volatility.

Sortino Ratio

Sortino Ratio improves upon Sharpe Ratio by considering only downside risk, making it more suitable for investment evaluation.

Alpha and Beta

Alpha measures excess return generated relative to benchmark expectations, while Beta measures sensitivity to market movements.

Maximum Drawdown

Maximum Drawdown identifies the largest decline experienced by a fund from its peak value.

Fund Scorecard

A composite scorecard was developed using multiple performance indicators to rank mutual funds objectively.

Key Performance Findings

- Funds with higher Sharpe Ratios generally delivered superior risk-adjusted returns.
- Alpha values helped identify funds outperforming benchmarks.
- Maximum Drawdown highlighted downside risk exposure.
- Top-ranked funds demonstrated strong returns with controlled volatility.

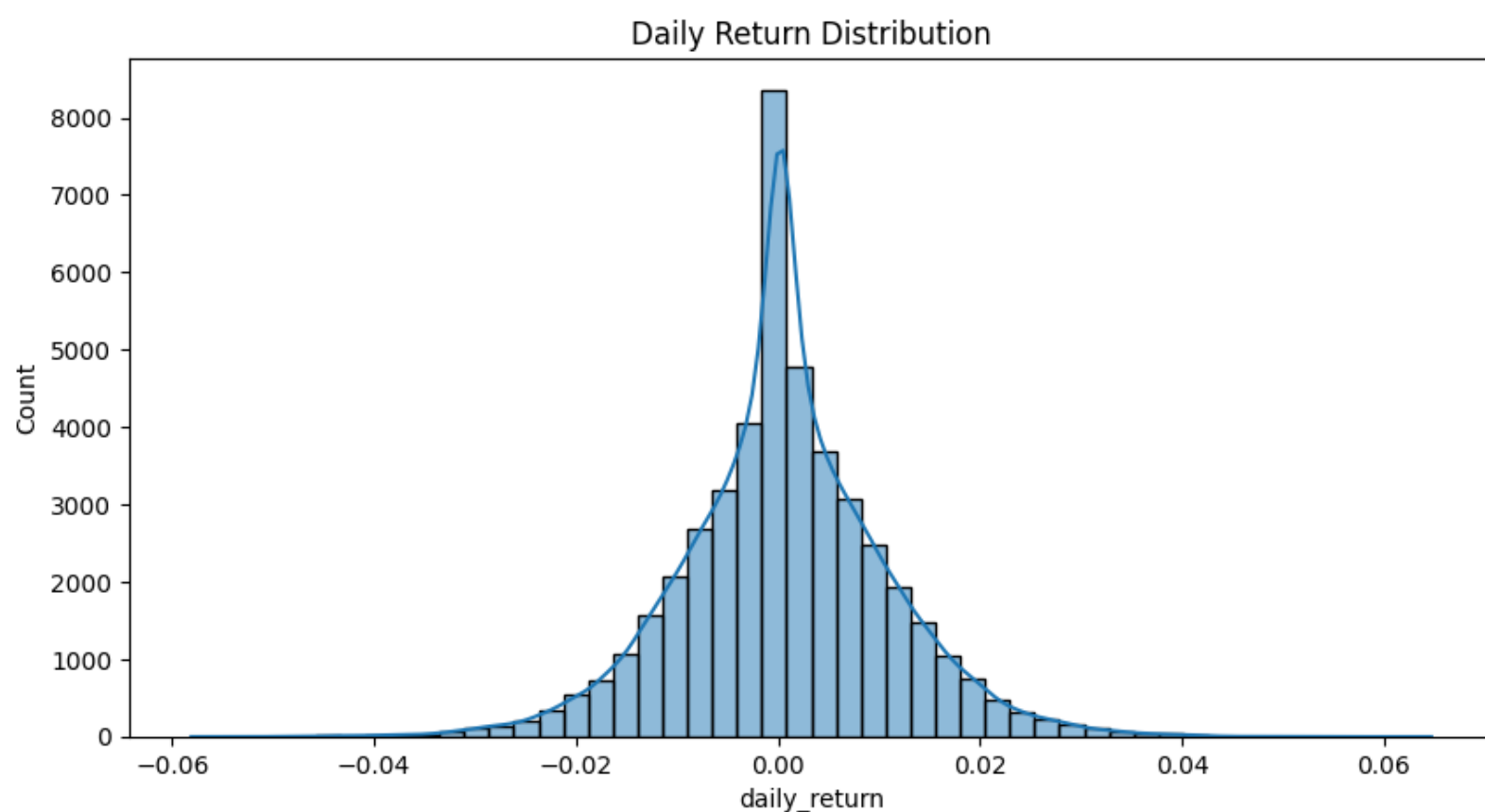


Figure 5: Daily Return Distribution

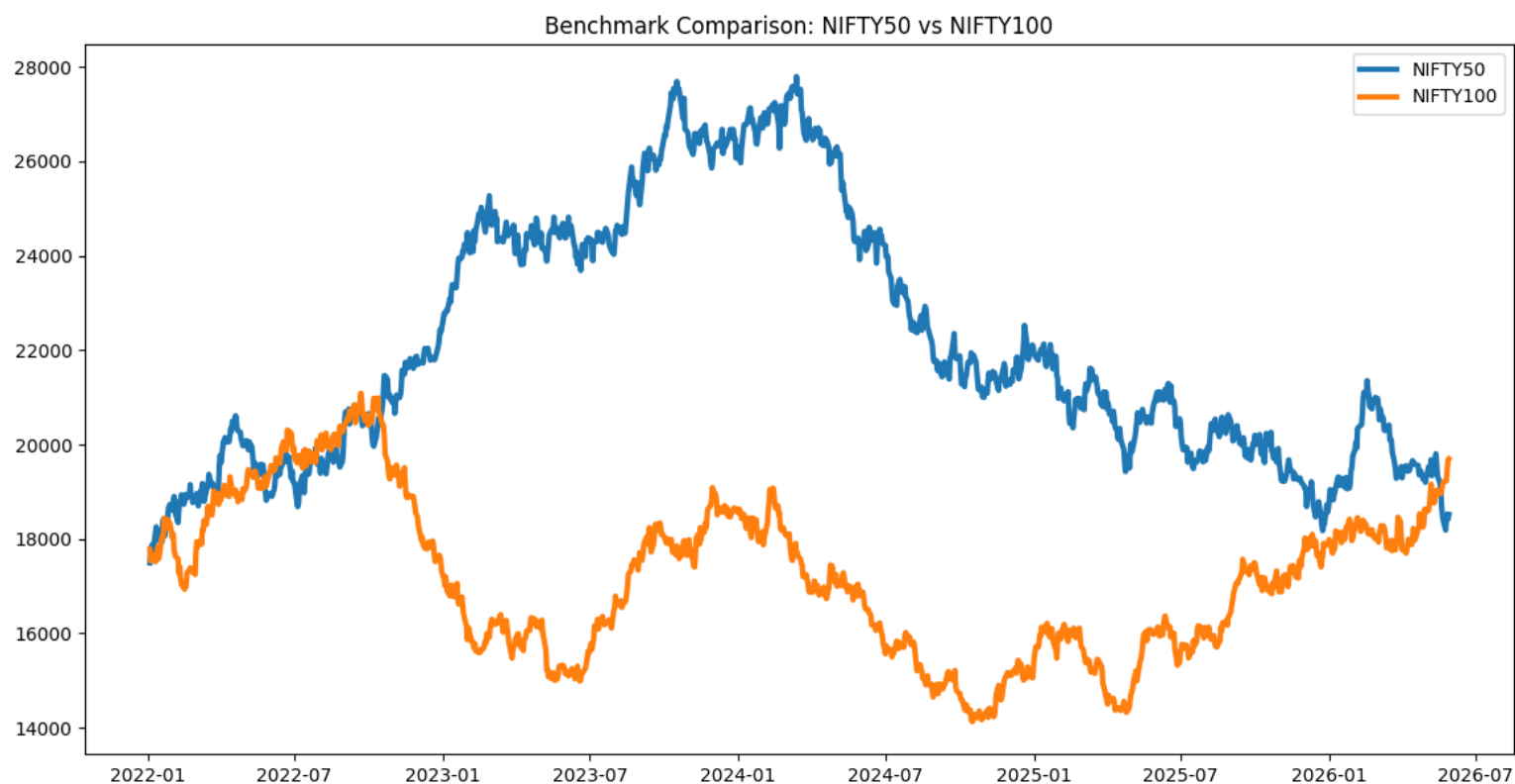


Figure 6: Benchmark Comparison Analysis

➤ Advanced Analytics:

Advanced analytics were performed to assess risk, investor behaviour, and portfolio concentration.

❖ Value at Risk (VaR);

Historical Value at Risk (95%) was calculated using daily return distributions. VaR estimates the maximum expected loss under normal market conditions at a specified confidence level.

❖ Conditional Value at Risk (CVaR);

CVaR measures the average loss beyond the VaR threshold and provides a better understanding of extreme downside risk.

❖ Rolling Sharpe Ratio;

A rolling 90-day Sharpe Ratio was calculated to monitor changes in risk-adjusted performance over time.

❖ Investor Cohort Analysis;

Investors were grouped based on their first transaction year to analyze long-term investment behaviour and participation patterns.

❖ SIP Continuity Analysis;

SIP transaction frequency was analyzed to identify investors at risk of discontinuing their SIP investments.

❖ Fund Recommendation System;

A rule-based recommendation engine was developed to recommend suitable mutual funds based on investor risk appetite.

❖ Sector Concentration Analysis;

The Herfindahl-Hirschman Index (HHI) was used to measure portfolio concentration across equity funds.

✓ Key Advanced Analytics Findings;

- High VaR funds exhibited greater downside risk.
- Funds with stable rolling Sharpe Ratios demonstrated consistent performance.
- Recent investor cohorts showed strong participation in mutual fund investments.
- Most investors maintained SIP discipline.
- Concentrated portfolios exhibited higher sector-specific risk.

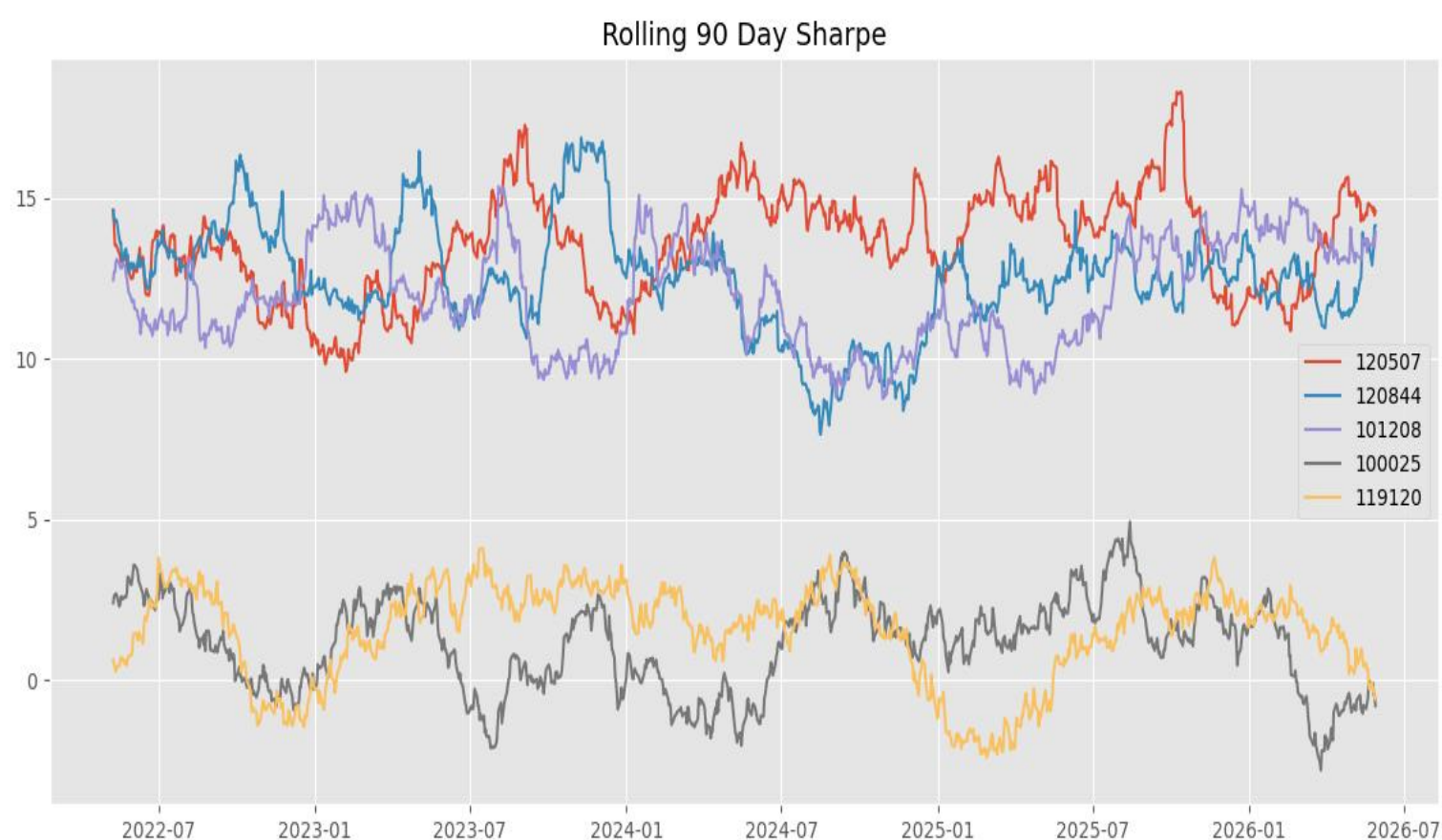


Figure 7: Rolling 90-Day Sharpe Ratio

➤ **Power BI Dashboard :**

An interactive Power BI dashboard was developed to provide business-friendly insights into mutual fund performance and investor behaviour.

The dashboard consists of four pages:

- **Page 1: Industry Overview**

Provides a high-level view of industry AUM, SIP inflows, folio counts, and scheme statistics.

- **Page 2: Fund Performance**

Displays fund returns, risk metrics, benchmark comparisons, and fund rankings.

- **Page 3: Investor Analytics**

Analyzes investor transactions, demographics, state-wise participation, and SIP behaviour.

- **Page 4: SIP and Market Trends**

Tracks SIP growth trends, category inflows, and benchmark market movements. Interactive slicers and drill-through functionality enable dynamic exploration of data.

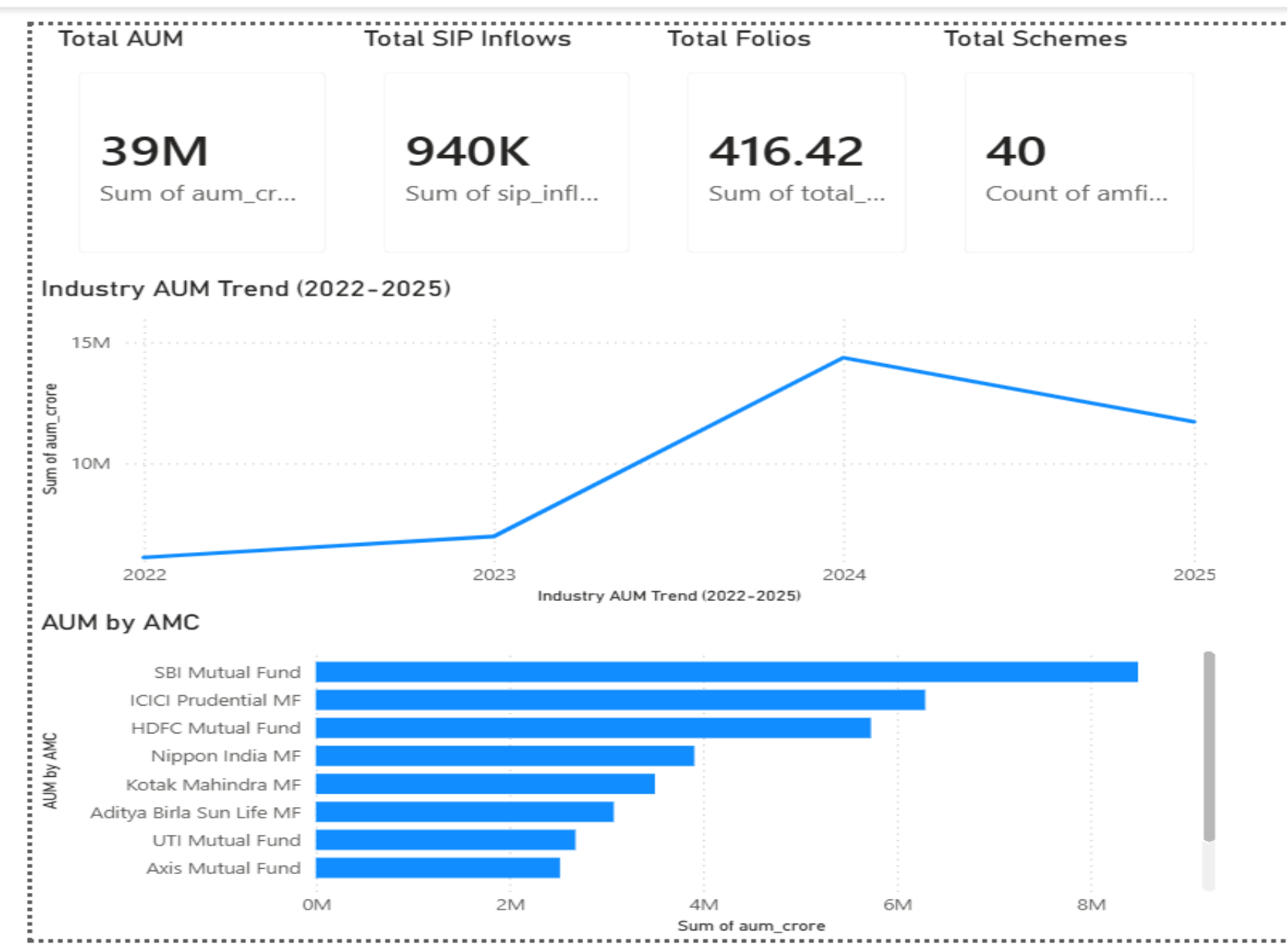


Figure 8: Industry Overview Dashboard

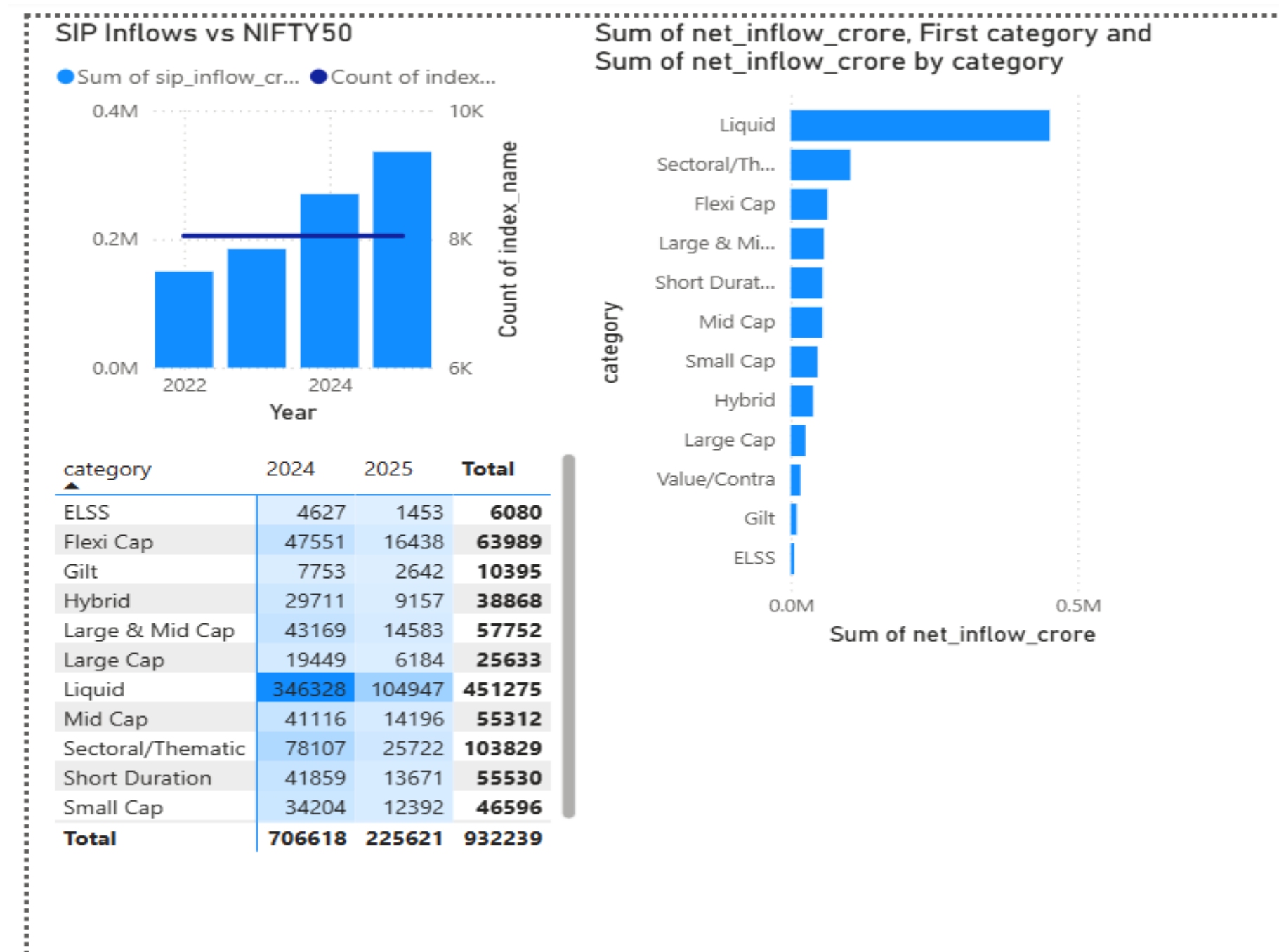


Figure 11: SIP & Market Trends Dashboard

✓ Key Findings :

1. Mutual fund industry AUM demonstrated strong long-term growth.
2. SIP inflows increased steadily, reflecting growing investor confidence.
3. Several funds consistently outperformed benchmark indices.
4. Risk-adjusted performance varied significantly across funds.
5. Younger investors showed higher SIP participation.
6. High HHI funds were more concentrated and potentially riskier.
7. Investor retention remained strong, with most SIP investors maintaining continuity.
8. Advanced analytics helped identify top-performing funds and risk exposure.

✓ **Limitations :**

1. Analysis depends on historical data and may not predict future performance.
2. Market conditions can change significantly over time.
3. Some datasets may contain assumptions or simulated values.
4. Fund recommendations are rule-based and not personalized financial advice.
5. External macroeconomic factors were not incorporated into predictive models.

✓ **Recommendations :**

1. Investors should evaluate both returns and risk before investing.
2. SIP investing should be encouraged for long-term wealth creation.
3. Portfolio diversification can reduce concentration risk.
4. Fund performance should be reviewed periodically.
5. Future versions can integrate real-time NAV APIs and predictive analytics.

✓ **Conclusion :**

The Bluestock Mutual Fund Analytics Capstone Project successfully developed an end-to-end analytics platform for mutual fund analysis. The project combined data engineering, database management, exploratory analysis, financial performance measurement, advanced analytics, and dashboard visualization into a single integrated solution.

The final system provides meaningful insights into mutual fund performance, investor behaviour, risk exposure, and industry trends. The project demonstrates the practical application of Python, SQL, SQLite, Power BI, and financial analytics techniques in solving real-world business problems.

❖ **Business Impact and Dashboard Benefits :**

The Power BI dashboard developed in this project provides a centralized platform for analyzing mutual fund performance, investor behaviour, and industry trends.

✓ Benefits for Investors;

- Easy comparison of mutual funds based on returns and risk.
- Better understanding of SIP trends and investment opportunities.
- Quick identification of top-performing funds.

✓ Benefits for Fund Houses;

- Improved visibility into investor participation.
- Better tracking of category-wise inflows.
- Data-driven decision making.

✓ Benefits for Analysts;

- Faster access to key performance indicators.
- Interactive filtering and drill-through analysis.
- Simplified reporting and visualization.

The dashboard transforms raw financial data into actionable business insights.

❖ Future Scope :

Several enhancements can be implemented in future versions of this project.

1. Integration with live NAV APIs for real-time updates.
2. Automated ETL scheduling using cron jobs or task schedulers.
3. Machine learning models for return prediction.
4. Portfolio optimization using Markowitz Efficient Frontier.
5. Monte Carlo simulation for long-term NAV forecasting.
6. Streamlit-based web application for interactive analytics.
7. Automated email reporting and alert generation.
8. Cloud deployment for enterprise-scale access.

These enhancements can significantly improve the analytical capabilities and business value of the platform.

❖ **Appendix :**

➤ **Technologies Used;**

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- SQLite
- SQL
- Power BI
- Git
- GitHub
- Jupyter Notebook

➤ **Project Deliverables;**

- ETL Pipeline
- SQLite Database
- EDA Notebook
- Performance Analytics Notebook
- Advanced Analytics Notebook
- Power BI Dashboard
- Final Report
- Presentation Slides

➤ **Key Output Files;**

- var_cvar_report.csv
- fund_scorecard.csv
- alpha_beta.csv
- tracking_error.csv
- rolling_sharpe_chart.png
- Dashboard.pdf
- bluestock_mf_dashboard.pbix